



# Cambridge (CIE) IGCSE Computer Science



Your notes

## SQL

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### What is SQL?

- **SQL** (Structured Query Language) is a programming language used to interact with a **DBMS**.
- The use of SQL allows a user to:
  - **Select data**
  - **Order data**
  - **Sum data**
  - **Count data**

### Selecting data commands

| Command       | Description                                           | Example                                                                                                                                                                             |
|---------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SELECT</b> | Retrieves data from a database table                  | <b>SELECT * FROM</b> users;<br>(retrieves all data from the 'users' table)<br><br><b>SELECT</b> name, age<br><b>FROM</b> users<br>(retrieves names and ages from the 'users' table) |
| <b>FROM</b>   | Specifies the tables to retrieve data from            | <b>SELECT</b> name, age <b>FROM</b> users;<br>(retrieves names and ages from the 'users' table)                                                                                     |
| <b>WHERE</b>  | Filters the data based on a specified condition       | <b>SELECT * FROM</b> users<br><b>WHERE</b> age > 30;<br>(Retrieves users older than 30)                                                                                             |
| <b>AND</b>    | Combines multiple conditions in a <b>WHERE</b> clause | <b>SELECT * FROM</b> users<br><b>WHERE</b> age > 18 <b>AND</b> city = 'New York';<br>(retrieves users older than 18 and from New York)                                              |



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|                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OR</b>        | Retrieves data when at least one of the conditions is true                                                                                                         | <b>SELECT * FROM</b> users<br><b>WHERE</b> age < 18 <b>OR</b> city = 'New York';<br>(retrieves users younger than 18 or from New York)                                                                                                                                                                                                                                   |
| <b>WILDCARDS</b> | '*' and '%' symbols are used for searching and matching data<br>'*' used to select all columns in a table<br>'%' used as a wildcard character in the LIKE operator | <b>SELECT * FROM</b> users;<br>(retrieves all columns for the 'users' table)<br><br><b>SELECT * FROM</b> users <b>WHERE</b> name <b>LIKE</b> 'J%';<br>(retrieves users whose names start with 'J')                                                                                                                                                                       |
| <b>ORDER BY</b>  | How data is organised (sorted) when it is retrieved                                                                                                                | <b>SELECT</b> Forename, Lastname <b>FROM</b> Students<br><b>WHERE</b> StudentID < 10<br><b>ORDER BY</b> Lastname, Forename <b>ASC</b><br><br>(retrieves only the <i>forename</i> and <i>lastname</i> of all students from the <i>students</i> table who have a <i>studentID</i> of less than 10 and displays in ascending order by <i>lastname</i> and <i>forename</i> ) |
| <b>SUM</b>       | Adds up and outputs the sum of a field                                                                                                                             | <b>SELECT SUM</b> (Salary) <b>FROM</b> tbl_people;                                                                                                                                                                                                                                                                                                                       |
| <b>COUNT</b>     | Counts the number of files which match the set criteria                                                                                                            | <b>SELECT COUNT</b> (*)<br><br><b>FROM</b> tbl_people<br><br><b>WHERE</b> Salary > 50000;                                                                                                                                                                                                                                                                                |

## Examples

- Select all the fields from the Customers table

Command:

```
SELECT * FROM Customers;
```

Output:



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| ID | Name      | Age | City     | Country |
|----|-----------|-----|----------|---------|
| 1  | John Doe  | 30  | New York | USA     |
| 2  | Jane Doe  | 25  | London   | UK      |
| 3  | Peter Lee | 40  | Paris    | France  |

- Select the ID, name & age of customers who are older than 25

Command:

```
SELECT ID, name, age
FROM Customers
WHERE Age > 25;
```

Output:

| ID | Name      | Age |
|----|-----------|-----|
| 1  | John Doe  | 30  |
| 3  | Peter Lee | 40  |

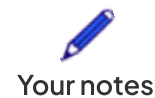
- Select the name and country of customers who are from a country that begins with 'U'

Command:

```
SELECT Name, Country
FROM Customers
WHERE Country LIKE 'U%';
```

Output:

| Name | Country |
|------|---------|
|------|---------|



|          |     |
|----------|-----|
| John Doe | USA |
| Jane Doe | UK  |

- Select all fields of customers who are from 'London' or 'Paris'

Command:

```
SELECT *  
FROM Customers  
WHERE City = 'London' OR City = 'Paris';
```

Output:

| ID | Name      | Age | City   | Country |
|----|-----------|-----|--------|---------|
| 2  | Jane Doe  | 25  | London | UK      |
| 3  | Peter Lee | 40  | Paris  | France  |



## Worked Example

Below is a table of animals called **tbl\_animals**

| Animal           | Breeding | Number of Young |
|------------------|----------|-----------------|
| Red Fox          | Yes      | 4–6             |
| Rabbit           | Yes      | 4–12            |
| African Elephant | Yes      | 1               |
| Blue Whale       | No       | 1               |
| Orangutan        | Yes      | 1               |
| Polar Bear       | Yes      | 1–3             |
| Dolphin          | Yes      | 1               |



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|          |     |     |
|----------|-----|-----|
| Kangaroo | Yes | 1   |
| Lion     | Yes | 1-6 |
| Penguin  | Yes | 1   |

Complete this SQL statement to display all of the animal breeds that are currently breeding and there was only one young born this year [3]

SELECT .....

FROM .....

WHERE .....

Answer

**SELECT** Animal [1]

**FROM** tbl\_animals [1]

**WHERE** Breeding = "Yes" **AND** Number of Young = 1 [1]